

Customer Retention and Churn Analysis Report

Data Science Intern Project

1 Introduction

This project analyzes customer churn behavior in a subscription-based telecom company. The objective is to understand why customers leave, identify key retention drivers, and evaluate customer lifetime value (CLV).

The analysis is presented through an interactive Streamlit dashboard: <https://futures01-dwqbshck6.streamlit.app/>

2 Dataset Description

The dataset used is the Telco Customer Churn dataset from Kaggle. It contains customer demographic information, subscription details, contract type, payment method, monthly charges, tenure, and churn status.

3 Methodology

The analysis was performed using Python and included:

- Data cleaning and preprocessing
- Churn rate analysis
- Retention analysis by contract type, tenure, and payment method
- Customer lifetime value calculation

4 Key Findings

- Overall churn rate is approximately 26.58%
- Month-to-month customers have the highest churn rate
- Electronic check users are most likely to churn
- Customers with longer tenure show strong retention
- Average customer lifetime value is approximately \$2,283

5 Visual Analysis

- Churn distribution analysis
- Contract-based churn comparison
- Payment method churn behavior
- Tenure-based retention trends
- Customer lifetime value distribution

(You can optionally insert screenshots from your dashboard here using `\includegraphics`)

6 Business Recommendations

- Encourage long-term contracts through discounts
- Improve onboarding for new customers
- Target high-risk customer groups with retention strategies
- Introduce loyalty programs for long-term customers

7 Conclusion

The analysis shows that contract type, tenure, and payment method are key drivers of churn. Businesses can significantly reduce churn by focusing on early-stage engagement and promoting long-term commitments.